



AI-Powered Smart Waste Segregation & Recycling Assistant

1M1B AI for Sustainability Virtual Internship

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EcoSort AI — System Workflow



Abstract

is a Python-based artificial intelligence and machine-learning prototype designed to support responsible waste segregation and recycling decisions. It accepts an everyday waste description, predicts a waste category using TF-IDF text features and Logistic Regression, and retrieves structured guidance and sustainability tips from a prototype dataset. An optional Gemini generative-AI layer can provide conversational guidance.

The project is primarily aligned with SDG 12: Responsible Consumption and Production. It demonstrates AI as a practical decision-support tool while explicitly considering fairness, transparency, ethics, privacy, safety, and local variation in waste-management rules.

1. Introduction

Waste segregation is an everyday sustainability challenge. People encounter plastic packaging, food scraps, paper, glass, metals, batteries, electronics, textiles and other materials, and these items can require different handling. Uncertainty about classification can make responsible disposal difficult.

provides a simple interface where a user can type an item such as “old mobile phone”, “banana peel” or “plastic bottle” and receive a category, material information, disposal guidance and an eco tip.

2. Problem Statement

How might we use AI to help people identify and properly segregate everyday waste so that waste management and responsible consumption can become more sustainable?

3. Objectives

- Develop a simple ML prototype for waste-category classification.
- Provide understandable disposal/recycling guidance for representative waste items.
- Encourage informed decisions around recycling, reuse, composting and special handling.
- Demonstrate a practical Python + AI sustainability application.
- Include responsible-AI safeguards and clearly communicate limitations.

4. SDG Alignment

Primary SDG: SDG 12 — Responsible Consumption and Production.

The project supports awareness and decision support around waste segregation, recycling, reuse and responsible disposal. It does not claim to have measured a direct SDG outcome; its contribution is designed as a prototype awareness and decision-support mechanism.

5. Target Users

- Households and families.
- Students and educational institutions.
- Offices and small organizations.
- Community members interested in sustainable living.

6. Proposed Solution

combines a machine-learning classifier with a structured knowledge dataset. The classifier predicts a category from the user's description. The knowledge layer then provides a material label, guidance type, disposal recommendation and eco tip.

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Example interaction

Input	Old mobile phone
Predicted category	E-waste
Material	Electronic
Guidance	Use an appropriate e-waste collection/recycling route.
Eco tip	Repair, reuse or responsibly recycle electronics.

7. Methodology

7.1 Data Preparation

The prototype contains 29 representative waste-item records. Each record contains an item description, material, category, disposal guidance and eco tip.

7.2 Text Representation

TF-IDF converts short natural-language descriptions into numerical features. Word and two-word n-grams allow the model to learn phrases such as “mobile phone” and “plastic bottle”.

7.3 Machine-Learning Classification

A Logistic Regression classifier is trained on the prototype examples and predicts one of the available waste categories for a new description.

7.4 Guidance Retrieval

After classification, the application retrieves a relevant structured record. This separates predictive classification from informational guidance and makes the guidance easier to review and update.

7.5 Optional Generative AI

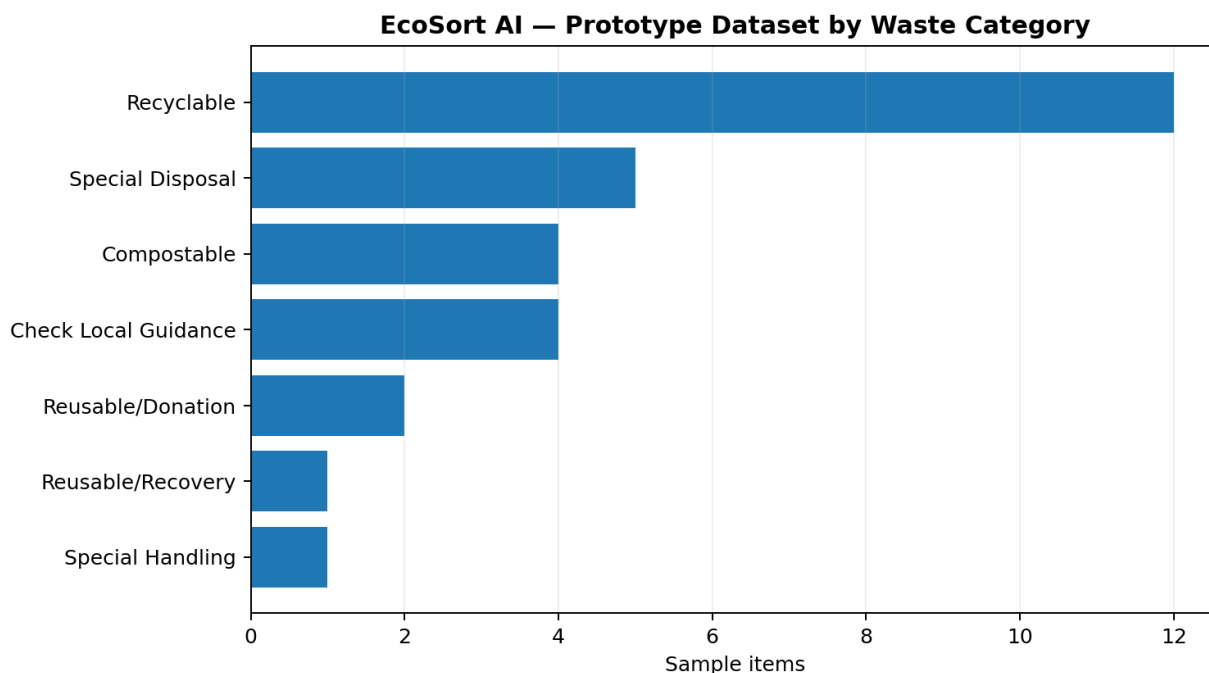
When a Gemini API key is configured, the system can generate a concise conversational explanation based on the predicted category and known record. The core application remains usable without an API key.

8. Technology Stack

Technology	Purpose
Python	Application logic
Streamlit	Interactive interface
Pandas	Dataset handling
Scikit-learn	Machine learning
TF-IDF	Text feature extraction
Logistic Regression	Classification
Gemini API (optional)	Generative AI guidance
Git/GitHub	Version control

9. Dataset & Analytics

The dataset is intentionally representative rather than a complete municipal waste database. Categories include Plastic, Organic, Paper, Glass, Metal, E-waste, Hazardous, Textile, Wood and Other.



The analytics view gives users a quick overview of the prototype data and can be extended as the dataset grows.

10. Responsible AI Considerations

- **Fairness:** The classifier does not use personal, demographic or sensitive attributes.
- **Transparency:** The application identifies its core model and separates prediction from structured guidance.
- **Ethics:** The system avoids presenting uncertain local rules as universal facts.
- **Privacy:** No personal information is required for the core workflow.
- **Safety:** Hazardous waste and e-waste are separated from ordinary household recycling categories.
- **Local context:** Users are reminded to verify official local waste-management instructions.

11. Testing Plan

Input	Expected category
Plastic bottle	Plastic
Banana peel	Organic
Newspaper	Paper
Old mobile phone	E-waste
Used battery	Hazardous
Old clothes	Textile

12. Expected Impact

- Make basic waste categories easier to understand.
- Support more informed everyday disposal decisions.
- Encourage recycling, reuse and composting where appropriate.
- Provide a scalable starting point for localized sustainability guidance.
- Demonstrate responsible AI applied to a real-world sustainability problem.

These are expected/planned impacts; the prototype does not claim measured environmental reduction.

13. Limitations

- The dataset is small and representative rather than exhaustive.
- Waste-management and recycling rules vary by location.
- Ambiguous descriptions can reduce classification reliability.
- Real deployment would require broader validation and authoritative local data.
- Generative-AI output requires verification and should not override official guidance.

14. Future Scope

- Image-based waste recognition.
- RAG using verified municipal guidance.
- Multilingual support.
- Local recycling and collection-point discovery.
- Mobile application.
- Feedback-based evaluation and model improvement.

15. Conclusion

demonstrates how Python, machine learning and optional generative AI can be combined to address an everyday sustainability problem. The project uses AI for classification and decision support while keeping the workflow understandable and responsible. Its design is aligned with SDG 12 and provides a clear foundation for future localization and multimodal expansion.

16. Project Deliverables

- Working Streamlit prototype and source code.
- Prototype waste dataset.
- Requirements file and GitHub-ready README.
- Detailed project report.
- Professional presentation.